Brian Sawaya

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Education

University of California, Berkeley Berkeley, CA
B.S. Mechanical Engineering Expected May 2028
Coursework: Experimentation & Measurements (LabVIEW, DAQ), Dynamic Systems & Feedback, Electronics for IoT

Skills

Design & Analysis: SolidWorks, Onshape, FEA, GD&T, tolerance stack-up, kinematic coupling, DFM/DFA, FMEA, V&V
Build & Test: 3D printing, manual mill, lathe, laser cutter, soldering, LabVIEW, DAQ, oscilloscope, strain gauges
Electronics & Software: Arduino, I2C, Bluetooth Low Energy, stepper motor control, Python, MATLAB, C++, AI agents

Experience

Samtec, *R&D Optics Engineer Co-op* Aug 2026 – Present
- Built a **VCSEL test station** that recreates component error states tied to field failures and customer issues
- Reverse-engineered a Suruga Seiki motorized positioner's controls and **automated VCSEL-to-VCSEL moves** in Python
- Designed 3D printed fixtures with kinematic locking and wrote CCD camera software to verify beam alignment
- Validated **≤30 µm positioning repeatability** with 30-sample tests (standard deviation, commanded vs. actual)

Boeing, *Supply Chain Engineering Intern* May 2026 – Aug 2026
- Built AI agents linked to supplier data sources that sorted 6,000+ entries, cutting **turnaround time by 50%**
- Streamlined Production Readiness Assessments used across 12,000+ suppliers, cutting **time per assessment ∼40%**
- Documented process observations on supplier site visits, used to score suppliers against **AS9100** requirements

JH Technology Associates LLC, *Systems Engineer Intern* Nov 2024 – May 2026
- Tested and integrated subsystems of a spectrophotometer now on the **International Space Station**, finding 35 error states
- Calibrated the device from MATLAB analysis of intensity and absorption data, raising **signal-to-noise ratio by 20%**
- Modeled wire management in SolidWorks, soldered components to PCBs, and 3D printed **45+ prototype parts**

Stanford University, *Research Intern, Structural Biology* Sep 2024 – Dec 2024
- Developed a skin adapter for an Optical Coherence Tomography scanner used for noninvasive biopsy and cancer monitoring
- Rapid-prototyped 7+ CAD revisions with kinematic coupling, reaching **4.6 µm Z-repeatability** vs. a 500 µm spec

Projects

Linear Array Detector for Mass Spectrometry, *Engineering Solutions @ Berkeley* Sep 2025 – Present
- Designing a proof-of-concept detector that identifies viral infections from vaporized saliva by mass-to-charge ratio
- Held **<0.1 mm positional tolerance** between all plates with reamed dowel pin holes and a resin-printed alignment jig
- Compared 9 manufacturing methods and reached **∼40% ion transmittance** across a 19-hole array (60% target)

AutoLab: Automated Cell Culturing Robot, *Project Lead, Neurotech @ Berkeley* Aug 2025 – Present
- Lead a team of 5 on a **$2,000 Jacobs Institute grant**, building a low-cost multi-axis gantry for cell culturing
- Designed a stepper-driven micropipette (A4988 drivers) and a purely mechanical tool changer for swapping culturing tools
- Achieved **0.1 mm gantry repeatability**; selected to present at a statewide neurotech conference

Industry Client Project (under NDA), *Engineering Solutions @ Berkeley* Sep 2025 – Dec 2025
- Evaluated manufacturability of a proposed assembly process for a medical client; delivered a **go/no-go recommendation**
- Ran **SolidWorks contact FEA** to predict part deformation and stress under fixture loads
- Authored the **design FMEA and process FMEA**, wrote a test protocol, and designed a joint concept in CAD

Diversity By Doing HealthTech, *Intern, Stanford Biodesign* Jan 2025 – Apr 2025
- Led a team of 5 researching endometriosis and wrote a clinical needs statement with **Intuitive Surgical** mentors

Leadership

MESA, *Ambassador* Feb 2024 – 2025
- Connected 105+ students to internships, raised **$2,000 in scholarships**, and ran events for 500+ people

Achievements

1st Place, UC Davis Biomedical Engineering Design Challenge (2025) **TEDx Speaker**, *"Engineering Your Luck"* (2025)